

Hyder Ali

*PhD Degree in Electronics and
Communication Engineering.
Anna University*

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Area of Interest (Teaching)

Electronics, Computer, Information, Communication, Digital Image and Signal Processing, Machine & Deep Learning.

Key Skills

- ❖ 30 years teaching experience as Professor – 4 years, Associate Professor – 4 years, Assistant Professor -10.3 years, Lecturer – 10.3 years, and Archer Cornfield Fellow – 1 year, in various departments. (ECE, EEE, IT, Computer Engineering).
- ❖ 4 years industrial experience as Electronics Engineer and Customer Support Engineer.
- ❖ Journal publications: 9, Conference publications: 6
- ❖ Accreditation process in NBA, NAAC, ABET, NCAAA.
- ❖ Programming skills in C, Python, MATLAB in the field of Image Processing.
- ❖ Research interest in Machine Learning and Deep Learning.
- ❖ Conducting conferences and workshops.
- ❖ Members in Professional societies like ISTE, IETE, IE(I), BESI, IACSIT, SDIWC, IAENG and SSI
- ❖ Attended and conducted several professional In-Service Training Programs.
- ❖ Proficiency in Information Technology and Learning Management Systems (LMS)

Educational Qualification

- 2008 - 2011 PhD Degree in Electronics and Communication, Anna University, India
Title: “Certain Investigations on Denoising Techniques for Brain MR Images Using Curvelet Transform” <https://shodhganga.inflibnet.ac.in/handle/10603/13371>
Defended on August 12, 2011
Laboratory: Signal and Image processing Lab, Thiyagarajar College of Engineering, Madurai.
Supervisor: Prof. Dr. R. Sukanesh, Thiyagarajar College of Engineering, Madurai
- 1997 - 1999 Master Degree in Communication Systems, Madurai Kamaraj University, India
Grade & Percentage: I class & 72.42%.
Defended on February 5, 1999
Title: Data acquisition and retrieval software for NI-VXI 1500 DAS using Lab Windows/CVI
Laboratory: Department of Data Acquisition, Liquid Propulsion System Center, Indian Space Research Organization, India, 1998
Supervisor: Prof. Dr. R. Sukanesh, Thiyagarajar College of Engineering, Madurai
- 1985 - 1989 Bachelor Degree in Electronics and Communication Engineering, Mohamed Sathak Engineering College/Madurai Kamaraj University, India
Grade & Percentage: I class & 71.58%

Professional Experience - Academia

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- 6/5/2025 – **Visiting Faculty in** Guilin University of Aerospace Technology, China
 - 1/7/2025 ○
 - 01/2024 – **Professor in** Electrical and Electronics Engineering, Ashesi University, Ghana.
 - 9/2022- ○ **Archer Cornfield Fellow** in Electrical and Electronics Engineering, Ashesi University,
 - 12/2023 Ghana.
 - 10/2014 - ○ **Assistant Professor** in College of Engineering & Information Technology, Buraydah
 - 05/2021 Colleges, Saudi Arabia.
 - 06/2013- ○ **Professor** in Electronics & Communication Engineering, RMD College of Engineering,
 - 09/2014 Chennai, India.
 - 09/2011- ○ **Professor** in Information Technology, RMK Engineering, College, Chennai, India
 - 05/2013
 - 08/2007- ○ **Associate Professor** in Electronics & Communication Engineering, AR College of
 - 09/2011 Engineering & Tech, Tenkasi, Tamil Nadu India
 - 04/2003- ○ **Assistant Professor**, in Electronics & Communication Engineering, Noorul Islam College of
 - 08/2007 Engineering, Nagercoil, Tamil Nadu, India
 - 06/1999 – ○ **Lecturer**, in Electronics & Communication Engineering, Noorul Islam College of
 - 03/2003 Engineering, Nagercoil, Tamil Nadu, India
 - 09/1991 – ○ **Lecturer**, in Electronics & Communication Engineering, Al-Ameen Institute of Technology,
 - 06/1997 Erode, Tamil Nadu, India
 - Industry**
 - 09/1997 –
 - 08/1999 ○ **Customer Support Engineer**, Mighty Electronics Equipment Corporation, Madurai Division
 - 07/1989- ○ **Electronic Engineer**, Mighty Electronics Equipment Corporation, Coimbatore, Tamil Nadu,
 - 08/1991 India

Courses Taught

- Electronics Devices and Circuit
- Signals and Systems
- Computer Architecture
- Circuit Theory
- Digital Image Processing
- Programming in C, Python and Tensor Flow
- Logic Design Lab
- Communication System Lab
- Electronics Devices and Circuit Lab
- Digital Electronics & Logic Design
- Digital Signal Processing
- Introduction to simulation and PCB Design
- Medical Image Processing.
- Microprocessor and Micro controller
- Machine Learning/Deep Learning
- Data Communication Lab
- DSP lab
- MATLAB Programming Lab

Technical and Linguistic skills

- Programming ○ C, Python
- Networking ○ NS3, CISCO Packet Tracer.
- Machine Learning ○ Tensor Flow, Keras API, PyTorch,
- Python ○ Data Science: Pandas, NumPy, Matplotlib, and Seaborn
- Libraries ○ Image Processing: OpenCV, SciPy, Scikit-image, Pillow
- IDE ○ PyCharm, Jupyter Notebook, Spyder
- Tools ○ MATLAB/Simulink, PSpice, Circuit Maker, LAB View, MS Office, Altium Designer
- Languages ○ English: Fluent, Tamil: Native, Arabic: Basic
- LMS ○ Talent MS, Moodle, Canvas.

Publications

International Journals (9)

- [1]. *Sampoornam K P., Dr. Hyder Ali Segu Mohamed., and Saranya S ***“IoT-based Smart Irrigation and Monitoring System for Agriculture”*** Computational Intelligence in Robotics and Automation (10.1201/9781003181668-9) (2022). 29(05), 161-186.
- [2]. *Dr. Hyder Ali Segu Mohamed., ***“A Social Bots Detection Based on Behavior Enhanced with Machine Learning Technique”*** International Journal of Advanced Science and Technology (Q4- Scopus Indexed) May 2020. Vol: 29(05), Page: 9229-9233. ISSN: 2005-4238 (Print) ISSN: 2207-6360 (Online).
- [3]. *K. P. Sampoornam, Hyder Ali Segu Mohamed, and R. Priyadarshini, ***“Economics Analysis of Amazon EC2 Instances”***, Journal of Computational and Theoretical Nano science, (Scopus Indexed) (EISSN: 1546-1963, 2018, 15(5), 1728–1733, ISSN: 1546-1955 (Print)
- [4]. Dr. Yosef Jbara, *Dr. Hyder Ali. S, ***“An efficient Artifact Free Denoising Technique for MR Images relying on Total Variation based Thresholding in Wavelet Domain”***, ICGST Journal of Graphics, Vision and Image Processing, ISSN 1687-398X, 2018, Volume 18, Issue 1, ICGST LLC, Delaware, USA, June
- [5]. Rajasekar. R, *Sasikumar.S, Hyder Ali. S, ***“Integrate Numerous Sampling Tasks In Sensor Field Using Geometric Algorithm”***, Journal of Signal Processing and Wireless Networks, 2016, Vol. 1, No. 1, pp 23-28,
- [6]. *Hyder Ali. S., Dr. Sukanesh. R, ***“An Effective Algorithm for Denoising MR and CT Images Using Digital Curvelet Transform”*** Springer Advances in Experimental Medicine and Biology – Software Tools and Algorithms for Biological Systems, ISSN: 0065-2598 2011, Volume 696, Part 6, pp 471-480, Plenum Press, USA (SCIE Indexed)
- [7]. *Hyder Ali. S., Dr. Sukanesh., ***“An Edge Preserving Denoising Technique for MR Images Using Curvelet Transform”***, The journal of Institution of Engineers (India), May 2010, Volume 91, pp 3-8.
- [8]. *Hyder Ali. S., Dr. Sukanesh., ***“An efficient Artifact Free Denoising for MR Images by Combining Wavelet and Total Variation”*** International Journal of Advanced Computer Engineering, ISSN: 0974-5785 2011, Volume 3, No 2, pp 97-103.
- [9]. *Hyder Ali. S, Dr. Sukanesh. R, ***“An Improved Denoising Algorithm using Curvelet Thresholding Technique for Medical Images”*** International Journal of Advanced Computing. ISSN: 0975-7686, April 2010, Volume 2, Issue 2 pp 83-91.

International Conferences (6)

- [10]. Yosef Hasan Fayez Jbara, *Hyder Ali Segu Mohamed, ***“Twitter Spammer Identification using URL based Detection”***, IOP Conference Series: Materials Science and Engineering, 2020, 925 012014
- [11]. *K. P. Sampoornam, Hyder Ali Segu Mohamed, R. Priyadarshini, ***“Economics Analysis of Amazon EC2 Instances”***, International Conference on Cyber Physical Systems for Next Generation Computing (CPSNGC-2018), India, March 1-3, 2018, 135-140
- [12]. *S. Hyder Ali, Dr.R. Sukanesh, Padma Priya, ***“Medical Image Denoising using Neural Networks”***, International Conference on Information and Automation (ICIA’05), Colombo, Sri Lanka, 15-18 2005, 102 – 107.
- [13]. *S. Hyder Ali, Dr.R. Sukanesh, R. Avinsha, ***“Automated Segmentation in Computer Tomography Imaging”***, International Conference on Mathematical and Engineering Techniques in Medical and Biological Sciences (METMBS’04), Las Vegas, USA. June 23-24, 2004, 320-328.
- [14]. *S. Hyder Ali, Dr.R. Sukanesh, Sathya, ***“Fusion of MRI and CT Using Wavelet Transform”***, Fourth National Conference on Advanced Computing (NCAC2004) PSG College of Technology, CSE, Coimbatore, India. December 10-11, 2004, 92-98.

- [15]. *Dr.R. Sukanesh, Krishnaswami, S. Hyder Ali, ***“Simplified coding method for progressive transmission of digital Images”***, National conference on Control Communication & Information System (CCIS’04), Goa Govt Engineering College, India January 23-24, 2004 73-77.

Journal and conferences review activities

- Medical & Biological Engineering & Computing, Springer
- Journal of Super Computing, Springer
- Journal of Pharmaceutical Research International
- International journal on Biomedical Engineering: Applications, Basis and Communications.
- SCIENCEDOMAIN internationals.
- International Advisory Committee, in international conference on “Computing and Communication”.
- ICSPS 2019: 11th International Conference on Signal Processing System

Professional Society Membership

- Life Member in Indian Society for Technical Education, India – LM 43835
- Life Member in Institute of Electronics and Telecommunication Engineers – F176093
- Life Member in Broadcast Engineers Society of India, India. – LM 1835
- Corporate Member in the Institution of Engineers (India) – M-139859-2
- Senior Member in International Association of Computer Science and Information Technology - 80343282
- Member in Society of Digital Information and Wireless Communication (SDIWC) – 1745
- Member in Engineering & Scientific Research Group (ESR Groups)
- Member in International Association of Engineers (IAENG) – 137043
- Member in Semiconductor Society of India (SSI)- 201401660

Professional Training - 11

- Completed certificate course in “Python Course on multiple languages” Conducted during October 2021, by Centre of Continuing Education, IIT Madras Digital Skills Academic Programs
- Training workshop on “Assessment of Program Learning Outcome” conducted by NCAAA on 13-14/11/2019
- Workshop on “NBA – New Outcome Based Accreditation”, 4 Days: 21.09.2013 & 26-28.09.2013
- Training on “Faculty Induction Training” 5 Days: 25-29 May 2009
- Certificate Course on “Source Coding of Speech, Images and Video”, 8 days, 31.03.’09 – 07.04.’09
- Seminar on ‘Medical Imaging & Processing’ 2 days 4, 5.03.2005
- Project training in “Assembly Programming In 8085” 1 month, 15.11.’04 – 15.12.’04
- Course Training in “Programming in C and C++” 3 months, 01.02.’04 – 30.04.’04 (PT)
- Course Training in “VLSI and Embedded Systems” 3 Days. 26 – 28 July 2002
- Bridge Course in “Digital Signal Processing”, 2 weeks: from 11 June to 23 June 2001.
- Workshop on “Training the trainers”, 2 Days: 16 – 17 July 1999

Webinar Attended - 9

- One Week Faculty Development Programme on "Digital VLSI Design and verification" Organized by Department of ECE, BIT, India, during 23rd – 27h August 2021
- One Week International FDP on "Emerging Trends & Technologies in Data Science" Organized by Department of CSE, GMR IT, India, during 24th – 29th August 2020

- One day webinar on “Feature Extraction Techniques in Medical Imaging” held on 9th July, 2020 organized by Department of Electronics and Communication Engineering, CMR Engineering College, Hyderabad-501401.
- Five days FDP on “IPRs for Professional Innovators” from 16th June to 20th June 2020, Organized by AVIT & Vinayaka Mission’s Research Foundation (VMRF).
- One day webinar on Applied Deep Learning, some of the concepts of Deep Learning like TensorFlow, PyTorch, Neural Networks, Deep Fake are discussed in Webinar. Held on 14-June-2020 organized by Indian Servers
- Three Days Online Faculty Development Programme on “IoT, AI & ML: Industrial Perspective” Organized by the Department of ECE, Sri Venkateswara College of Engineering and Technology (Autonomous), Chittoor held on 5 th June 2020 - 7th June 2020.
- Webinar on “Emerging Technologies & Use of AI in Banking and Financial Services” presented by Mr. Rajesh Vasudevan, Vice President, Maveric Systems Ltd., Singapore on 4th June 2020, organized by Department of ECE, SVCET Chittoor, Andhra Pradesh.
- Two days Online Faculty Development Program on Cyber Security organized by Department of Computer Science and Engineering of Swarnandhra College of Engineering & Technology during 15th & 16th May 2020
- One Week Faculty Development Program on Online Teaching Etiquette and Best Practices from 10-05-2020 to 15-05-2020, Organized by Talentio Solutions India Pvt. Ltd., Hyderabad

Students’ supervision

- Predictive Pest Management: A Machine Learning-Based Electronic Scarecrow.
- EMG Based Control for assisting devices, using machine learning.
- Speech Analysis using Wavelet Transform
- FFT based Spectrum Analyzer using TMS320C50.
- Digital Image Enhancement Compression using Wavelet Transform.
- Image filtering using median and averaging filters.
- Medical Image Fusion using Wavelet Transform.
- Wavelet Denoising of Medical Images
- Automated Segmentation in Computer Tomography Imaging.
- Medical Image Denoising using Neural Network
- Multiresolution Watermarking Using Wavelet Transform

References

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